

National Academy of Science and Technology

(Affiliated to Pokhara University)

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First Terminal Examination

Level: Bachelor

Semester : VI_Spring

Year : 2025

Program: B.E. Computer

F.M. : 100

Course: Machine Learning

P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Machine Learning. Discuss its evolution and how it differs from traditional programming approaches. 7
- b) Explain the different types of Machine Learning with suitable examples. Describe the typical Machine Learning workflow. 8
2. a) Explain the working of Linear Regression. How do you interpret its coefficients? 8
- b) What are regularization techniques? Compare L1 and L2 regularization with examples. 7
3. a) The table provides a set of eight training examples of the target concept "flu", where each instance is described by the attributes *Chills*, *Runny Nose*, *Headache* and *Fever*. Construct the decision tree for the "flu", using information gain. Show each steps and the final tree. 10

Chills	Runny Nose	Headache	Fever	Flu
Yes	No	Mild	Yes	No
Yes	Yes	No	No	Yes
Yes	No	Strong	Yes	Yes
No	Yes	Mild	Yes	Yes
No	No	No	No	No
No	Yes	Strong	Yes	Yes
No	Yes	Strong	No	No
Yes	Yes	Mild	Yes	Yes

- b) Differentiate between supervised and unsupervised learning with examples. 5
4. a) Explain the working of the Support Vector Machine (SVM) algorithm. Include the concept of margin and kernel trick. 8
- b) Explain the difference between regression and classification metrics. Provide examples of each. 7
5. a) Describe different model validation techniques. Explain k-fold cross-validation. 8

OR

What is a confusion matrix? How does f1 score differ from accuracy model evaluation.

- b) What are the major challenges in Machine Learning? Explain how they impact model development. 7
6. a) Apply K (=2)-Means algorithm over the data (185, 72), (170, 56), (168, 60), (179, 68), (182, 72), (188, 77) up to two iterations and show the clusters. Initially choose first two objects as initial centroids, Centroid for first cluster $c_1 = (185, 72)$ and Centroid for second cluster $c_2 = (170, 56)$. 7

OR

What is K-Means clustering? Describe the steps involved and how the optimal number of clusters is determined.

- b) Differentiate between classification and regression problems in supervised learning. Provide one use case for each. 8
7. Write short notes on: (Any two) 2×5
- a) Hyperparameter
- b) Reinforcement and Active Learning
- c) Bias-Variance Tradeoff